

The empirical recurrence for OEIS A316180, recovered from its tail

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Abstract

OEIS A316180 records “Empirical recurrence of order 74” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 130$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 80$. Its last coefficient is $c_{74} = 20 \neq 0$, so no further tail satisfies anything shorter; any order-74 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A316180 is “Number of $n \times 5$ 0..1 arrays with every element unequal to 0, 1, 2, 3, 4 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

16, 420, 2665, 21825, 168051, 1361115, 11056918, 90607484, ...

The entry states:

Empirical recurrence of order 74 (see link above)

As of the “Last modified” line on the live entry (Jun 25 2018 17:47:25, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 130$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 130$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 74: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 74.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 260$ exact terms from $n = 6$ on, it returns order 74 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{74} c_i a(n-i),$$

c_1	7	c_{20}	67114245	c_{39}	749493885	c_{58}	47381268
c_2	38	c_{21}	8881429	c_{40}	1792467251	c_{59}	−11582812
c_3	−163	c_{22}	−181530142	c_{41}	−545285173	c_{60}	−13428505
c_4	−788	c_{23}	−9881189	c_{42}	−1703983522	c_{61}	1399467
c_5	1219	c_{24}	390248896	c_{43}	262075077	c_{62}	1857729
c_6	7721	c_{25}	−30594194	c_{44}	1574239809	c_{63}	1852738
c_7	−1600	c_{26}	−685771245	c_{45}	−190243243	c_{64}	−416974
c_8	−44898	c_{27}	133719025	c_{46}	−1665623774	c_{65}	−898419
c_9	−37411	c_{28}	1031561080	c_{47}	474149109	c_{66}	151619
c_{10}	167807	c_{29}	−360374868	c_{48}	1453728086	c_{67}	237710
c_{11}	321031	c_{30}	−1320765999	c_{49}	−681632673	c_{68}	−46757
c_{12}	−339987	c_{31}	800833449	c_{50}	−855368941	c_{69}	−36473
c_{13}	−1421942	c_{32}	1351644735	c_{51}	611322889	c_{70}	7382
c_{14}	−151895	c_{33}	−1296528776	c_{52}	242649414	c_{71}	3280
c_{15}	3975489	c_{34}	−1117700152	c_{53}	−339997605	c_{72}	−734
c_{16}	4019442	c_{35}	1412947603	c_{54}	78070974	c_{73}	−84
c_{17}	−6346423	c_{36}	1023581706	c_{55}	82193016	c_{74}	20
c_{18}	−19451745	c_{37}	−1071265350	c_{56}	−105679681		
c_{19}	2645089	c_{38}	−1401650788	c_{57}	13134192		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 80$, and it is the recurrence OEIS A316180 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{74} - c_1 x^{74-1} - \dots - c_{74}$. Its constant term is $-c_{74} = -20$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of

terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 74 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 74, so $\deg q = 74$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 74, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 74, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 20.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-74 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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